Blockchain & Bitcoin

Distributed Ledger Technology (DLT) also known as the Blockchain enables transactions to be recorded on a decentralized digital ledger. Bitcoin was first implemented buy Satoshi Nakamoto as bitcoin was the first cryptocurrency to transact on the blockchain. Blockchain is one type of DLT and Bitcoin is a specific application of Blockchain technology for the use of payment tokens. Most cryptocurrencies use blockchain as the basis for their transactions. A blockchain network compiles and records peer-to-peer transactions using primarily Proof-of-Work or Proof-of-Stake consensus mechanisms to validate transactions. Once the transaction is validated, cryptocurrency transactions are recorded by adding blocks of code in a linear manner to the ever-expanding ledger of the blockchain network.

Public blockchain networks include Bitcoin and Ethereum. The transaction amount, timestamp, and wallet address of the sender and recipient are publicly available. Wallet balances are also publicly available. **www.etherscan.io** displays Ethereum wallets and transactions, including ERC20 tokens and NFTs minted on the Ethereum Blockchain. **www.blockchain.com** allows the public to view bitcoin wallets and transactions. Blockchain networks can also be partially or completely private. Monero is known for its private network as the public cannot view wallet balances or transactions whatsoever.

An emerging type of distributed ledger is called a hashgraph which is a different type of DLT than blockchain. Transactions are recorded on a directed acyclic graph, which makes use use of vertices and edges in contrast to blockchain. As a result of this difference, hashgraphs have the potential to increase transaction speed and reduce energy usage. A hashgraph is also decentralized like blockchain, but it does not require miners as many cryptocurrencies do. Hedera Hashgraphis the pioneer of the haashgraph approach to a distributed ledger. Hedera Hashgraph's native token, HBAR, is a cryptocurrency that records transactions using a hashgraph rather than blockchain. Beyond recording peer-to-peer transactions, distributed ledger offer enormous potential to enhance data storage, power decentralized applications, optimize finance, enhance cybersecurity, and transform the internet as we know it.

Ethereum Ledger: https://etherscan.io/
Bitcoin Ledger: https://www.blockchain.com/

Smart Contracts

A smart contract is a digital contract that can be supported by certain blockchain networks such as Ethereum, Cardano, etc.. Smart Contracts remove the need for a third-party intermediary. The terms of a smart contract are coded into the block chain that the smart contract runs on. The contract's terms are available to view on public blockchain networks like Ethereum & Polkadot. Smart contracts self-terminate once the terms of the contract are met with out any third-party interaction needed. Smart contract termination requires gas fees paid using the native token of the smart contract.

For example, terminating Ethereum based smart contracts requires gas fees to be paid in ETH. Gas fees are payments made by users to compensate for the computing energy required to process and validate transactions on the Ethereum blockchain. A higher gas limit means that you must do more work to execute a transaction using ETH or a smart contract. Decentralized crypto exchanges use smart contracts to allows users to swap one token for another. The contract terms include the conversion rate of the two tokens and the amount to be sent and received between two parties. The smart contract replaces a traditional intermediary such as a centralized exchange and the smart contract code alone facilitates the exchange of cryptocurrencies.

Fungible Tokens

A fungible token is an asset that can be transferred on a blockchain network by paying gas fees in the native token of the blockchain network it is traded on. Fungible Tokens are sets of tokens that are identical, in contrast to non-fungible tokens (aka NFT's) which are all unique from each other. Each fungible token has its own smart contract address which is used to identify the fungible token on their respective blockchain network.

A few examples of blockchain networks that support fungible tokens are Ethereum, Binance Smart Chain, and Tron. They are known as ERC-20 tokens on the Ethereum Network, TRC-20 tokens on the Tron Network, and BEP-20 tokens on the Binance Smart Chain. As each token within the group is identical to each other, many cryptocurrency projects issue their tokens in this manner.

The ERC-20 tokens with the largest market capitalizations are Uniswap, Chainlink, Polygon and Shiba Inu. Some projects intend to remain permanently as ERC-20 tokens running on the Ethereum Network. Other projects, such as Holochain which is currently traded as an ERC-20 token, intend to develop their own blockchain and the ERC-20 token functions as an I-O-U for their blickchain's token in the future.

Decentralized exchanges are peer-to-peer trading platforms. Users can trade fungible and non-fungible tokens, along with the native token of the blockchain network they run on. Uniswap supports the trading of ETH and ERC-20 tokens. Users pay gas fees in ETH to execute smart contracts needed to transact on the exchange. Beyond trading on decentralized exchanges, Fungible Tokens have significantly expanded the ecosystems of the blockchains they are built on, such as the Ethereum Network and Binance Smart Chain.

Proof-of-Work & Proof-of-Stake

Proof-of-Work and Proof-of-Stake consensus mechanisms are different methods used to validate transactions on various blockchain networks. These consensus mechanisms are the driving force of mining and staking rewards. The rewards provided to those who validate transactions affect the cryptocurrency's inflation rate as reward distribution increases the cryptocurrency's circulating supply.

Blockchain networks that utilize the Proof-of-Work consensus mechanism, such as bitcoin, use a network of miners that provide computing power to validate transactions on its decentralized network. To incentivize miners to validate transactions they are rewarded with the native token of the respective blockchain network. Miners rewards inflate the circulating supply according to a predefined schedule. Miners typically receive a higher reward in the earlier years of cryptocurrency's existence because of halving events. Halving events usually occur every few years and result in the reduction of miner rewards by 50%. The most recent Bitcoin halving event is scheduled for March 4th 2024.

The Proof-of-Stake consensus mechanism utilizes designated validators to validate transactions instead of miners. To qualify as a validator, an individual must self-bond a significant amount of their own cryptocurrency of the respective blockchain network to their validator pool. Nominators can elect to delegate their position of cryptocurrency to a validators pool as well. There are far fewer validators on networks using Proof-of-Stake consensus mechanisms in comparison to the number of miners validating transactions on blockchain networks utilizing the Proof-of-Work consensus mechanism. Validators also use equipment that requires significantly less energy to validate transactions when compared to the energy needed to power miners' equipment. As a result, blockchain networks that utilize the Proof-of-Stake consensus mechanism generally have a faster transaction speed, lower transaction cost, and lower carbon footprint than blockchain networks using Proof-of-Work consensus mechanism.

Validators and their nominators are rewarded with periodic payments of the staked asset upon successful validation of transactions. The staking reward amount varies by blockchain network. The reward rate depends on the overall proportion of the reward compared to nominators. Validators and their nominators can be penalized if the validator behaves 'badly' while validating transactions efficiently and correctly. A notable upcoming event is the transition from Ethereum to Ethereum 2.0. Ethereum 2. will utilize a Proof-of-Stake consensus mechanism once it fully transitions away from Ethereum's current Proof-of-Work consensus mechanism. If Ethereum is launched successfully, it is estimated that the networks energy usage will be reduced by about 99% compared to it's current usage.